GitHub Research Report: Deep Third-Cumulant Cost Wall

ARC White-Box Estimation, Phase 2 geometry: width n=1024, depth L=16, budget B=2^41 FLOPs

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Search scope. I resumed from the GitHub API rate-limit point and continued with direct GitHub code/repository/commit search plus indexed GitHub web search. The search covered challenge-specific fingerprints (whest, K3_SIMPLE, FactoredTensor, D21, V29), Khatri-Rao/MTTKRP, Hadamard low-rank algorithms, TensorSketch, tensor hypercontraction/ISDF, tensor trains, dynamic low rank, polynomial/Wiener chaos, response/adjoint methods, and higher-cumulant implementations. I then opened the most relevant estimator files and experiment ledgers rather than relying on README titles.

Bottom line. I did not find a published exact representation that collapses V29's 107-u old-source tier to <=8 u/layer while retaining the measured 2% D21 tolerance. I did find one strong, immediately testable accuracy lever that appears absent from public V29: radial/angular reparameterization of the Gaussian input. I also found a new, sharply falsifiable cost direction inspired by ISDF/tensor hypercontraction: compress the already-contracted old-source query in a shared row space, not the Hadamard product.

1. The problem, identified precisely

The hard part is no longer K3 accuracy in the abstract. It is a **representation-cost mismatch**: the deep K3 information is empirically indispensable, but its natural source representation forces dense Hadamard-product contractions. The leaders' bill says that the same information must either be represented without source identity, or contracted through a shared object that avoids per-source dense re-formation.

```
For an old source b: A_b = Q F_A,b, P_b = Q F_P,b D21_b contains (A_b o P_b) A_b^T and (A_b o A_b) P_b^T.
Measured: A_b, P_b are low-rank; A_b o P_b and A_b o A_b are near full-rank.
```

The public V29 solution therefore uses the only obvious compromise: store old legs in shared low-rank bases, re-form enough dense content to take the Hadamard products, and pay approximately 107 u for the old tier. Your measurements close the usual alternatives: lower shared rank, CP/Tucker caps, source dropping, slice recursion, even-statistic closures, TensorSketch, and backward sensitivity.

What survived the GitHub audit

Finding	Status	Estimated cost	Why it matters
Angular/radial state	Strong - measured in a public Phase-2 K3 estimator	~0 extra u recurrent; O(n^2) first-layer work	Raw K3 MSE improved 7.9% at n=1024,L=16 in an independent public implementation.
Shared query-space ISDF / row skeleton	Conjecture - not yet measured on V29	~4-6 u/layer for old tier if shared dimension s<=256	Targets the rank~230 contracted query, not the full-rank Hadamard product.
Penultimate exact K4 residual contraction	Small accuracy rider	~1.02 u once with level-5 Strassen, if residual slices already available	Public code gives an exact two-matmul output correction; historical gain ~0.9%.
Shared squared-propagator identity	Derived negative	Exact lift is O(n^2 r^2), too expensive	Shows why T_b=T_c X does not by itself remove the Khatri-Rao wall.

2. Finding #1 - Angular/radial reparameterization

Evidence source: public repository `barnobarno666/ARC-White-Box-Estimation-2026`, Phase 8 report and `candidates/estimator_p8_final.py`. The repo records an official Phase-8 submission (#330178) at 69.58% utilization. More importantly for us, its local n=1024,L=16 ablation isolates the angular representation itself.

[Derived] Because the network has no biases and ReLU is positively homogeneous, every layer is degree-1 homogeneous in the input. Write a Gaussian input as

```
X = (R / sqrt(n)) Y, R = ||X||_2 ~ chi_n, Y = sqrt(n) X / ||X||_2, Y uniform on the sphere of radius sqrt(n), R independent of Y.
```

Then for every layer l, exactly:

```
h_l(X) = (R / sqrt(n)) h_l(Y).
```

For any homogeneous degree-p raw moment, the radial factor is known analytically:

```
a_p = E[(R/sqrt(n))^p] = 2^(p/2) Gamma((n+p)/2) / (Gamma(n/2) n^(p/2)). At n=1024: a1=0.99975589, a2=1, a3~1.00073233, a4=1.00195312.
```

The spherical input Y has Cov(Y)=I and zero third cumulant, but a nonzero fourth cumulant. If c4 denotes the diagonal fourth cumulant,

```
c4 = kappa4(Y_i,Y_i,Y_i,Y_i) = -6/(n+2) = -0.0058479532 at n=1024.
```

For first-layer preactivation covariance M=W W^T, the fourth-cumulant slices needed by a K3-SIMPLE style closure are

```
s4_i = c4 * M_ii^2 s22_ij = (c4/3) * (M_ii M_jj + 2 M_ij^2), i != j.
```

[Measured in the public Phase-8 code] On width 1024, depth 16: Gaussian K3 raw MSE 3.7859e-8 -> angular K3 3.4859e-8, a 7.92% reduction with 8/8 wins. On a locked 12-network confirmation panel the reduction was about 7.47% with 11/12 wins. K2 and K2+K4 variants improved 15.6-17.0%.

Why this is outside the V29 dead list. I searched the public V29 repo (`504aldo/whest-p2-cumulant-k3`) for angular/spherical/chi-state initialization. Its "radial" terminology refers to the K4 radial core, not this input-distribution reparameterization. Public V29 initializes mu=0, C=I and starts K4 state later; it does not appear to run the spherical-input chain.

Cost. The recurrent equations need not change. The added work is one first-layer O(n^2) construction for s22 plus O(n) output rescaling. In your unit D=n^2(2n-1), this is effectively <0.001 u/network beyond work V29 already performs.

Expected gain if it transfers. A naive multiplicative extrapolation gives 2.13e-8 x 0.9208 = 1.96e-8. Do not treat that as a prediction; V29 already has richer K4 regeneration and corrections, so overlap can reduce the gain.

First falsifier (one network, CPU). Add only (i) c4=-6/(n+2) and the first-layer s4/s22 injection, (ii) the corresponding spherical-state raw-moment bookkeeping, and (iii) multiply reported means by a1. Run the same stored network with 1e9 truth. Kill the idea if raw final MSE improves by <2%; promote to an 8-network ablation if it improves by >=4%.

Second experiment with possible cost payoff. If angularization helps accuracy, repeat only the old-tier rank cliff around the current operating point: R_OLD 384 -> 352 -> 320 and nested 224 -> 208 -> 192. The hypothesis is not that propagator rank magically drops; it is that removing radial closure error may buy enough total error margin to tolerate a lower old-tier approximation.

3. Finding #2 - Contract first, then compress: a query-space ISDF test

Inspiration from GitHub. Several numerical-linear-algebra repositories implement interpolative separable density fitting (ISDF), tensor hypercontraction (THC), or exact Khatri-Rao leverage sampling. The common idea is to avoid storing a rank-squared pair-product object and instead select/interpolate only the rows needed by the final contraction. Relevant public code includes OpenFermion ISDF/THC, `Yasuaki-Ito/GANSU`, and exact Khatri-Rao leverage samplers.

> **The key difference from the dead low-rank-Hadamard route: do not approximate A o P. Your M3 says A o P is near full rank, but the already contracted query U has a much smaller numerical rank: roughly 100 for 99% energy and about 230 for 2% error. That is the only low-dimensional object left in the pipeline.**

For an old source with A=Q F_A and P=Q F_P, one selected physical row i can be evaluated without forming dense A or P:

```
a_i = q_i^T F_A (length n) p_i = q_i^T F_P (length n) row_i(U) = (a_i o p_i o d)^T F_R^T
[channel-dependent right factor].
```

Thus s selected rows cost O(s r n) per source and channel instead of O(n^2 r) to form every row. If a shared left basis Q_U of dimension s reconstructs the aggregate query, only those rows are required.

Production conjecture. Reuse a query left-subspace from the preceding layer (or an anchor old source), transport/refresh it cheaply, evaluate only interpolation rows of the new aggregate old query, and reconstruct the full n x r query. This is not guaranteed by M3: M3 proves each/aggregate U is compressible, not that its left subspace is predictable without computing U.

FLOP estimate. A conservative four-channel row-evaluation model is approximately 12 s n r FLOPs per old source. In units D=n^2(2n-1):

s	r	row evaluation / source	one n x s by s x r reconstruction
224	224	0.287 u	0.048 u
256	224	0.328 u	0.055 u
256	384	0.563 u	0.094 u
384	384	0.844 u	0.141 u

A mixed layer with roughly ten old sources, mostly in the r=224 tier and a few in r=384, would land around **4-6 u/layer** if s<=256 and one shared query basis/reconstruction serves the aggregate. That is inside the requested 8-u/layer target. If s is 384 or the basis must be rebuilt from a full exact query, the advantage largely disappears.

First falsifier (do this before coding ISDF). At layers 10 and 14, save the exact old-tier contracted query matrices for all four channels. Normalize channels, then test three bases of dimension s=128,192,224,256,320: (1) oracle left SVD of the current aggregate query, (2) basis learned from one anchor old source, (3) previous-layer basis transported by W and re-orthogonalized. Measure max and RMS error for each individual channel, not only the sum.

```
PASS gate: s <= 256 AND every required channel <= 2% relative RMS, with an anchor/previous-layer basis
(not only oracle current-U SVD). KILL gate: oracle s > 256, or non-oracle reuse > 2% on any important
channel.
```

> **This experiment is cheap: it operates on stored n x r query matrices and randomized SVD/QR. It should finish on one 1024x16 network well under an hour on CPU. If it fails, the ISDF/THC analogy is irrelevant to this problem despite being excellent for chemistry pair densities.**

4. Why the shared squared propagator does not give an exact cheap identity

Your suggested algebra T_b = T_c X can be written exactly in a quadratic lift. Let S=T_c. For one birth-specific X,

```
[(S X) o (S X)]_ij = sum_{a,b} S_ia S_ib X_aj X_bj.
```

Define Phi(S) to contain all symmetric quadratic row features S_ia S_ib, with q=r(r+1)/2 features when S has rank r. Then

```
(S X) o (S X) = Phi(S) Psi(X) [exact].
```

[Derived negative] This shares the squared propagator across births, but it does not reduce arithmetic enough. For a generic Q in R^{n x r}, Phi(Q) has row rank min(n, r(r+1)/2). At r=224 or 384 this is generically n=1024, so there is no n-independent closure. A dense multiplication through the quadratic lift costs O(n^2 r^2).

```
At n=1024, r=224: n^2 r^2 / [n^2(2n-1)] ~= r^2/(2n) ~= 24.5 u before source-specific right contractions.
```

Therefore the identity is useful only if Phi(Q) has an unusually small *interpolative* rank for the particular contracted queries. That collapses exactly to the query-space experiment in Section 3. This is why I would not implement a full quadratic lift.

5. Hermite/Wiener chaos in the current W: exact decomposition, no hidden cheap order

For fixed incoming K3 and current Gaussian linear map B=W diag(w1), a D21 term is cubic in B:

```
D21_ik = sum_{a,b,c} B_ia B_ib B_kc K_abc.
```

For a Gaussian row, decompose

```
B_ia B_ib = Sigma_ab + H2_i,ab.
```

So the current-weight dependence terminates exactly into a degree-1 chaos term plus a degree-3 chaos term. The degree-1 term is the partial-trace / mean component already represented by the thin V29 machinery. The required fluctuation is precisely the H2 x B degree-3 part.

[Derived + matched to your measurements] There is no omitted degree-2, degree-4, etc. Hermite ladder to truncate. T2's 12-16% expectation-only floor says the degree-3 residual is far too large to discard. Compressing its coefficient tensor brings us back to the same Khatri-Rao/full-rank wall. Therefore generic polynomial-chaos GitHub libraries do not create a new state here.

6. Backward/adjoint route: an independent public negative result

The Phase-8 public repo also implemented direct frozen-source contraction with backward suffix matrices. Terminal-only contraction was cheap (~10% utilization) but inaccurate (raw MSE around 3.2e-6 even with the angular variant). All-depth direct contraction cost ~64.5% utilization and still produced roughly 4.65e-7 due to open-loop nonlinear drift. This independently matches your own adjoint/sensitivity and late-reformation measurements.

7. A cheap higher-cumulant rider found in another public GitHub project

Evidence source: `paulrosu11/arc-cumulant-mlp-estimator`, a repository containing hundreds of estimator experiments. Its geometry is older/smaller in several experiments, so the transfer is not assumed. The useful part is an exact contraction identity for penultimate K4 residual slices.

If s31 and s22 are the (3,1) and (2,2) K4 slice residuals after subtracting the cheap rank-2/K4 model at the penultimate nonlinear layer, their exact contribution to the final preactivation K4 diagonal is

```
delta4_i = 4 sum_{a,b} W[a,i]^3 W[b,i] s31[a,b] + 3 sum_{a,b} W[a,i]^2 W[b,i]^2 s22[a,b].
```

It can be evaluated with two matrix products:

```
G31 = s31 @ W term31_i = 4 sum_a W[a,i]^3 G31[a,i] G22 = s22 @ (W o W) term22_i = 3 sum_a W[a,i]^2 G22[a,i].
```

Cost at n=1024. Two dense matmuls are 2 u classically; with the allowed level-5 Strassen figure (~0.51 u/product), approximately **1.02 u once at the final layer**, plus O(n^2) elementwise work. In that public repo the additive exact residual improved raw MSE by only ~0.88%, so this is an accuracy rider, not a replacement for deep K3.

Important catch. V29 does not explicitly carry exact K4 residual slices; generating them naively would destroy the cost advantage. The experiment is therefore diagnostic first: obtain s31/s22 from an offline augmented teacher for one stored network, evaluate the final exact correction, and ask how much of V29's remaining mean error it explains.

```
PROMOTE if: final raw MSE improves >=5% on the one-network diagnostic. DEPRIORITIZE if: improvement <1-2%.
```

The same repo also uses very-low-rank symmetric CP/TT-like K4 channels whose W-contracted (2,1,1) diagonal costs O(n^2 r):

```
A_q = (W o W)^T (U_q o U_q), B_q = W^T U_q, delta4_i = 12 sum_q lambda_q A_iq B_iq^2.
```

At r=16-32 this is effectively free compared with an n^3 unit. It is worth testing only if a rank-32 fit captures a large fraction of the residual output correction; do not use it as an excuse to materialize K4 sources.

8. Public participant archaeology: what it tells us about the leaders

I found multiple Phase-2 participant repositories and recent commit histories. They are useful controls, but none exposes the present 1.64e-8 to 1.9e-8 at 133-165 u frontier.

Repository / artifact	What it demonstrates	Relevance to V29 wall
504aldo/whest-p2-cumulant-k3 (V29)	Public factorized K3, shared old bases, fused two contractions, K4 regeneration, Strassen engineering.	Our baseline; confirms wall and dead ends.
barnobarno666/... Phase 8	Angular state gave ~7.9% K3 MSE reduction; official score 2.3053e-8 at 69.58% utilization.	Strong accuracy trick, but still not the 133-165 u leader mechanism.
saademad200/arc-whitebox	K3_SIMPLE reached historical leader-class raw accuracy only when essentially uncapped; A=B symmetry cut generic factor cost.	A=B symmetry is already inherent in V29's shared propagator structure.
paulrosu11/arc-cumulant-mlp-estimator	Large experiment archive; K4 Tucker/TT/exact residual output corrections.	Useful for cheap K4 riders, not deep K3 cost.

Inference. The present leaders' deep representation is still probably private/unpublished. Exact score-string and compute-signature searches did not surface a matching public implementation. The public record supports the view that ordinary uncapped K3 can reach the right accuracy class but not the cost class; the missing innovation remains a deep-source contraction/state change.

9. Recommended experiment order

1 - Angular V29 patch

Cost: ~0 additional u. **Why first:** already measured positive at the exact n=1024,L=16 geometry in an independent K3 chain. **Kill:** <2% raw-MSE gain on one 1e9-truth network.

2 - Angular rank-cliff recheck

Cost to test: a handful of one-network runs. **Goal:** see whether the accuracy margin permits R_OLD/R_OLD2 reduction. This is the only immediate path from the angular accuracy gain to the 133-165 u cost regime.

3 - Shared query-space SVD/ISDF diagnostic

Cost: offline CPU minutes. **Pass:** s<=256 and <=2% error using a non-oracle reusable basis. If it passes, implement interpolation-row contractions; projected old-tier cost ~4-6 u/layer.

4 - Penultimate exact K4 residual oracle

Cost: ~1.02 u at final layer once integrated, but first test offline. **Pass:** >=5% raw-MSE gain from teacher residual slices. Otherwise ignore.

Decision rule after these four tests

If angularization gives its expected accuracy gain but the query-space reusable basis fails, then the GitHub search has not uncovered a credible <=8-u/layer deep-old representation. At that point I would stop searching generic tensor libraries: the remaining evidence says the leader mechanism is either unpublished, challenge-specific, or exploits a property not visible in per-source spectra. The next measurement should then be a production-grade temporal observability rank of the exact aggregate old query under the actual next-layer map.

10. Source index

G1 - 504aldo/whest-p2-cumulant-k3
`https://github.com/504aldo/whest-p2-cumulant-k3`
Public V29 implementation, findings log, generators and ablations.

G2 - barnobarno666/ARC-White-Box-Estimation-2026
`https://github.com/barnobarno666/ARC-White-Box-Estimation-2026`
Phase-2 experiment history; Phase 8 angular-state estimator/report and official submission record.

G3 - saademad200/arc-whitebox
`https://github.com/saademad200/arc-whitebox`
Independent K3_SIMPLE implementation, cost experiments, A=B invariant and rank-growth measurements.

G4 - paulrosu11/arc-cumulant-mlp-estimator
`https://github.com/paulrosu11/arc-cumulant-mlp-estimator`
Large archive of cumulant estimator experiments; exact K4 output residual and TT/Tucker channels.

G5 - quantumlib/OpenFermion ISDF/THC
`https://github.com/quantumlib/OpenFermion`
Interpolative separable density fitting and tensor hypercontraction implementations.

G6 - Yasuaki-Ito/GANSU
`https://github.com/Yasuaki-Ito/GANSU`
Least-squares tensor hypercontraction; randomized SVD on Khatri-Rao pair-product matrices.

G7 - vbharadwaj-bk/fast_tensor_leverage
`https://github.com/vbharadwaj-bk/fast_tensor_leverage`
Exact leverage-score sampling for large Khatri-Rao products without explicitly forming the full product.

G8 - tensorly/tensorly
`https://github.com/tensorly/tensorly`
Implicit MTTKRP and randomized CP routines; useful arithmetic reference but not an asymptotic escape for the dense-random V29 products.

G9 - tgcsaba/GPSig
`https://github.com/tgcsaba/GPSig`
Randomized low-rank Hadamard-product routines; rejected here by the measured flat Hadamard spectra.

G10 - oseledets/ttpy
`https://github.com/oseledets/ttpy`
Tensor train, TT-cross, rounding, dynamical low-rank references; largely closed by your pooled TT/Tucker rank measurements.

Research conclusion. The best concrete thing found on GitHub is the angular/spherical-input reparameterization: it is cheap, mathematically exact at the representation level, independently measured at the correct Phase-2 shape, and apparently absent from public V29. The best new cost experiment is not another tensor rank cap; it is a direct test of whether the *contracted old query* has a reusable <=256-dimensional row space. If that fails, I do not currently see a credible public method that beats the old-tier cost bound without violating your measured 2% D21 requirement.